



AI Smart Inventory Management System

An intelligent MERN-stack platform for inventory, GST billing & AI-driven business insights

Presented by: _____ | Roll No: _____

Guided by: _____ | Academic Year: 2025 – 26

The Problem

- Millions of small retail businesses (kirana stores, traders, wholesalers) still manage stock with **paper registers or basic spreadsheets**.
- Manual billing leads to **GST calculation errors**, lost invoices, and compliance headaches.
- Customer credit ("udhaar") is tracked in **physical khata books** — entries get lost, disputes arise, and payment reminders are forgotten.
- Shop owners have **no visibility** into which products sell fast, what to reorder, or how much stock is lost to shrinkage.
- Existing software is either **too expensive, too complex, or not built** for the Indian small-business workflow.

Project Objectives

Digitise Inventory

Replace manual registers with real-time stock tracking, units of measure, low-stock detection and shrinkage (stock-adjustment) records.

Simplify GST Billing

One-tap "Quick Sale" billing with automatic GST computation, PDF invoices and UPI QR codes for instant payment.

Add Intelligence

Use AI (Google Gemini) to forecast demand, generate reorder suggestions and surface actionable business insights.

Digital Khata

Track customer credit and repayments digitally — a ledger that never loses an entry.

Analytics & Reports

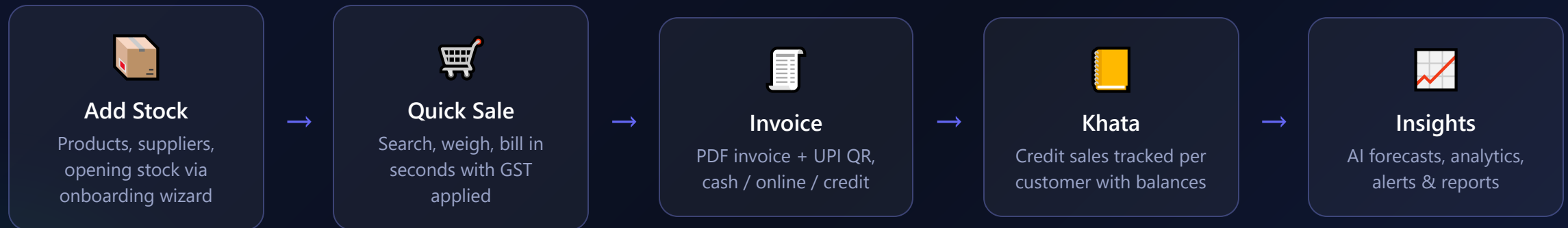
Dashboards, sales trends, automated scheduled reports and Tally-compatible export for accountants.

Accessible to All

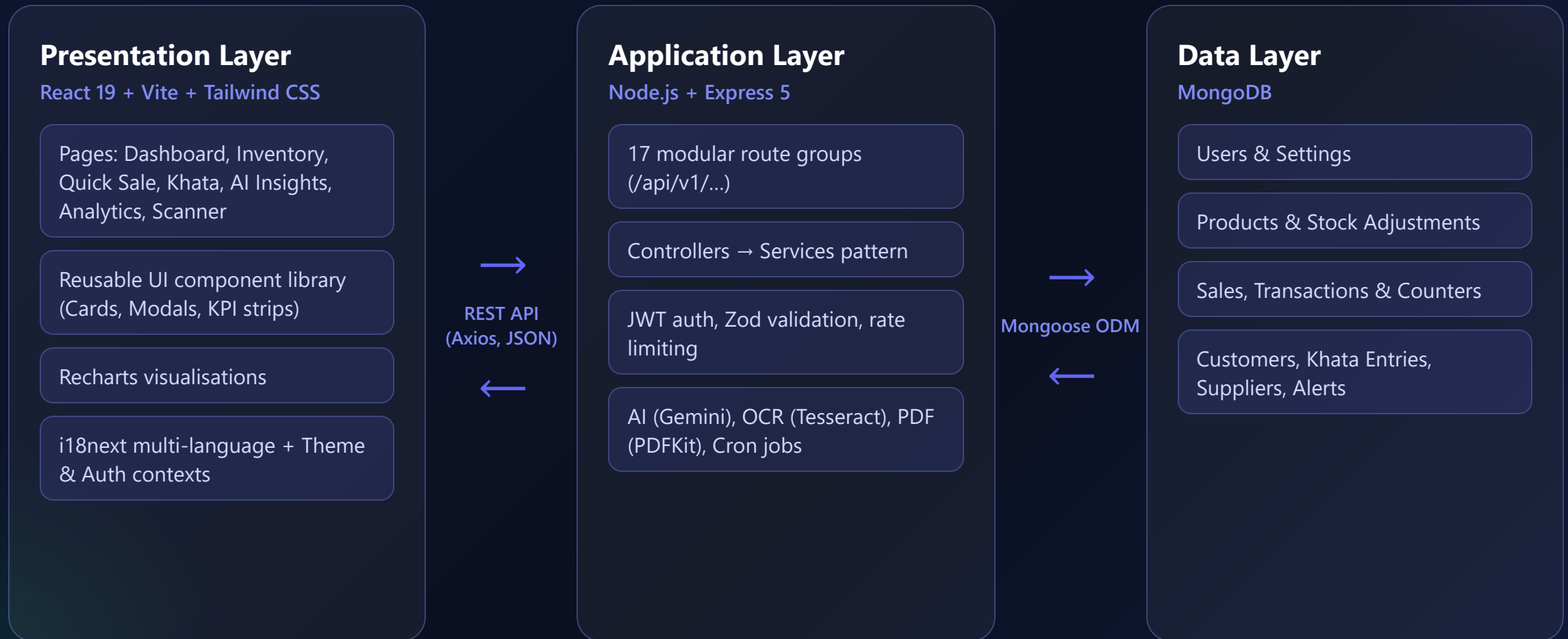
Multi-language interface (i18n), guided onboarding wizard, and a simple UI designed for non-technical shop owners.

Proposed System

- A **full-stack web application** built on the MERN stack — MongoDB, Express.js, React, Node.js.
- Single platform covering the complete retail workflow: **stock in** → **sell** → **bill** → **collect** → **analyse**.
- **AI-assisted decisions:** demand forecasting, smart alerts and an in-app help chatbot.
- **Automation:** cron jobs generate periodic reports and raise smart stock alerts without user effort.



System Architecture — 3-Tier MERN



Technology Stack

Layer	Technology	Purpose
Frontend	React 19, Vite, Tailwind CSS	Fast, responsive single-page application
Charts & UI	Recharts, Lucide icons, i18next	Analytics visualisations & multi-language UI
Backend	Node.js, Express 5	REST API with modular MVC + services architecture
Database	MongoDB + Mongoose 9	Flexible document storage, Decimal128 for money
AI	Google Gemini API	Demand forecasting, insights, help chatbot
OCR	Tesseract.js	Scan supplier bills to auto-extract products
Documents	PDFKit + QRCode	GST invoices with UPI payment QR codes
Security	JWT, bcrypt, Zod, express-rate-limit	Authentication, validation & abuse protection
Automation	node-cron	Scheduled smart alerts & report generation

Key Modules

Authentication

Register / login with JWT tokens, hashed passwords, guided 7-step onboarding wizard.

Inventory

Products with units of measure, stock levels, low-stock flags, shrinkage tracking.

Quick Sale & Billing

Amount-first or weight-first billing, GST computation, PDF invoice + UPI QR.

Customers & Khata

Customer directory with digital credit ledger and running balances.

Suppliers

Supplier records, purchase transactions, sample-pack shrinkage reports.

AI Insights

Gemini-powered demand forecasting, reorder report and help chatbot.

OCR Scanner

Upload a bill photo — Tesseract extracts items to add stock quickly.

Analytics & Reports

KPI dashboard, sales trends, scheduled reports, Tally export.

Smart Alerts

Cron-driven alerts for low stock, expiry and unusual activity.

Inventory & Product Management

- **Product catalogue** with categories, pricing, GST rate and unit of measure (pieces, kg, grams, litres).
- **Real-time stock levels** updated automatically on every sale and purchase.
- **Low-stock detection** with configurable reorder thresholds per product.
- **Stock adjustments** module records shrinkage — damage, theft, expiry, counting errors — with reasons.
- **Precise quantities:** weight stored with dedicated utilities to avoid floating-point errors.

Why it matters

The shop owner always knows **what is in stock, what is running out, and where stock is being lost** — replacing guesswork with data.

Implementation highlights

- Mongoose **Product** & **StockAdjustment** models
- Zod-validated REST endpoints
- Money & weight stored as **Decimal128** for accuracy
- Shrinkage page visualises losses by reason

Quick Sale & GST Billing

- **Designed for speed:** search a product, enter weight or amount, and the bill builds itself.
- **Amount-first mode:** customer says "₹50 of sugar" — the system back-calculates the quantity.
- **Tare-weight support** for weighing with containers.
- **Automatic GST** calculation per line item based on each product's tax rate.
- **Payment modes:** cash, UPI / online, or credit (goes straight to the customer's khata).
- **PDF invoice** generated server-side with PDFKit, including a **UPI QR code** for instant payment.



Digital Khata — Customer Credit Ledger

- Replaces the traditional paper **khata (udhaar)** book used in Indian shops.
- Every credit sale automatically creates a **khata entry** against the customer.
- **Running balance** maintained per customer — credits and repayments in one timeline.
- Repayments can be recorded **partially or fully**, with full history preserved.
- Owner sees **total outstanding credit** across all customers at a glance.

Data model

Customer ← linked to → **KhataEntry** (type: credit / payment, amount, linked sale, note, date) with a service layer that keeps balances consistent.

Impact

No more lost entries or disputes — both the shopkeeper and the data agree on **exactly who owes what**.

AI Insights & Smart Alerts

- **Demand forecasting:** sales history is analysed via the **Google Gemini API** to predict upcoming demand per product.
- **Reorder report:** AI suggests what to restock, how much, and when — before items run out.
- **Help chatbot:** an in-app assistant answers "how do I..." questions for non-technical users.
- **Smart alerts cron:** a scheduled job scans inventory and raises alerts for low stock and anomalies.
- **Demand chart** visualises forecast vs. actual sales using Recharts.

 Sales & stock history (MongoDB)



 ai.service.js → Gemini API prompt



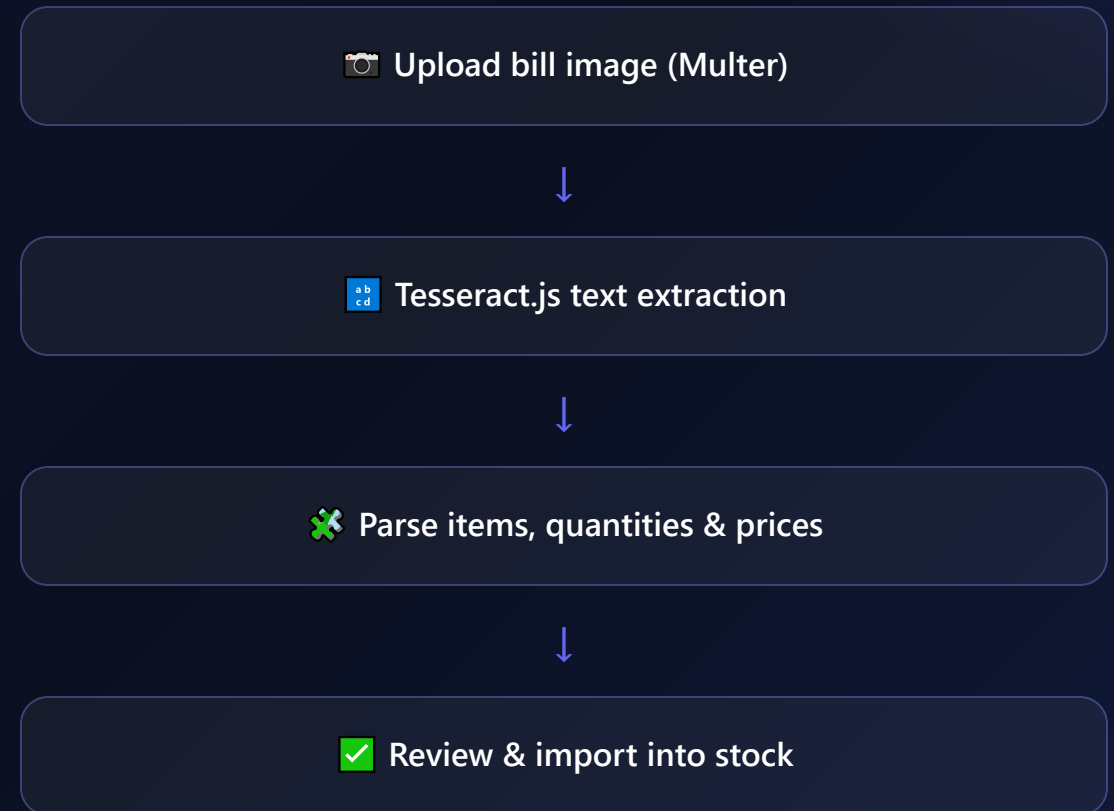
 Forecast + reorder suggestions (JSON)



 AI Insights page — charts & action cards

OCR Bill Scanner

- Adding 50 products from a supplier bill by hand is slow and error-prone.
- The owner simply **photographs the bill and uploads it** on the Scanner page.
- **Tesseract.js** performs optical character recognition on the server.
- The OCR service **parses product names, quantities and prices** from the extracted text.
- Parsed items are presented for **review and one-click import** into inventory.



Analytics, Reports & Integrations

- **Dashboard KPIs:** today's sales, outstanding khata, low-stock count and top products at a glance.
- **Analytics page:** sales trends, category performance and payment-mode breakdown rendered with Recharts.
- **Automated reports:** a cron job (reportGenerator) compiles periodic business reports without manual effort.
- **Tally export:** transaction data can be exported in a Tally-compatible format for the accountant.
- **Shrinkage & reorder reports** turn raw data into decisions: what is being lost, what to buy next.

17

API route modules

12+

Database models

15+

Frontend pages

2

Automated cron jobs

3+

Languages (i18n)

Security & Reliability

Authentication

JWT-based sessions with bcrypt-hashed passwords; protected routes via auth middleware on both API and UI.

Input Validation

Every endpoint validated with **Zod schemas** through a central validation middleware — malformed data never reaches the database.

Rate Limiting

express-rate-limit protects login and API endpoints from brute-force and abuse.

Money Accuracy

All currency and weight values use **Decimal128 / decimal.js** — no floating-point rounding errors in bills or GST.

Testing

Node test-runner smoke tests and end-to-end API tests verify endpoints, status codes and GST math.

Error Handling

Central error middleware on the server and React Error Boundaries with toast feedback on the client.

Future Scope

- **Mobile app** (React Native / PWA) so billing works at the counter on any phone.
- **WhatsApp / SMS integration** for sending invoices and automatic khata payment reminders.
- **Barcode scanning** at the billing counter for packaged goods.
- **Multi-store / multi-user workspaces** with role-based permissions for staff.
- **Offline-first mode** with sync — billing must never stop when the internet does.
- **Deeper AI:** price optimisation, festival-season demand models and anomaly detection for theft patterns.

Conclusion

- The system successfully **digitises the complete daily workflow** of a small retail business — inventory, billing, credit and analysis — in one application.
- **AI integration is practical, not decorative:** forecasts and alerts directly answer "what should I reorder and when?"
- The **MERN architecture** with a clean controllers → services → models structure makes the system modular, testable and easy to extend.
- Careful engineering choices — **Decimal128 money math, Zod validation, JWT security, cron automation** — make it reliable enough for real shops, not just a demo.
- The project demonstrates end-to-end full-stack skills: **database design, REST APIs, modern React, AI/OCR integration and automated testing.**



Thank You

Questions & Discussion

AI Smart Inventory Management System — MERN Stack Project, 2025–26